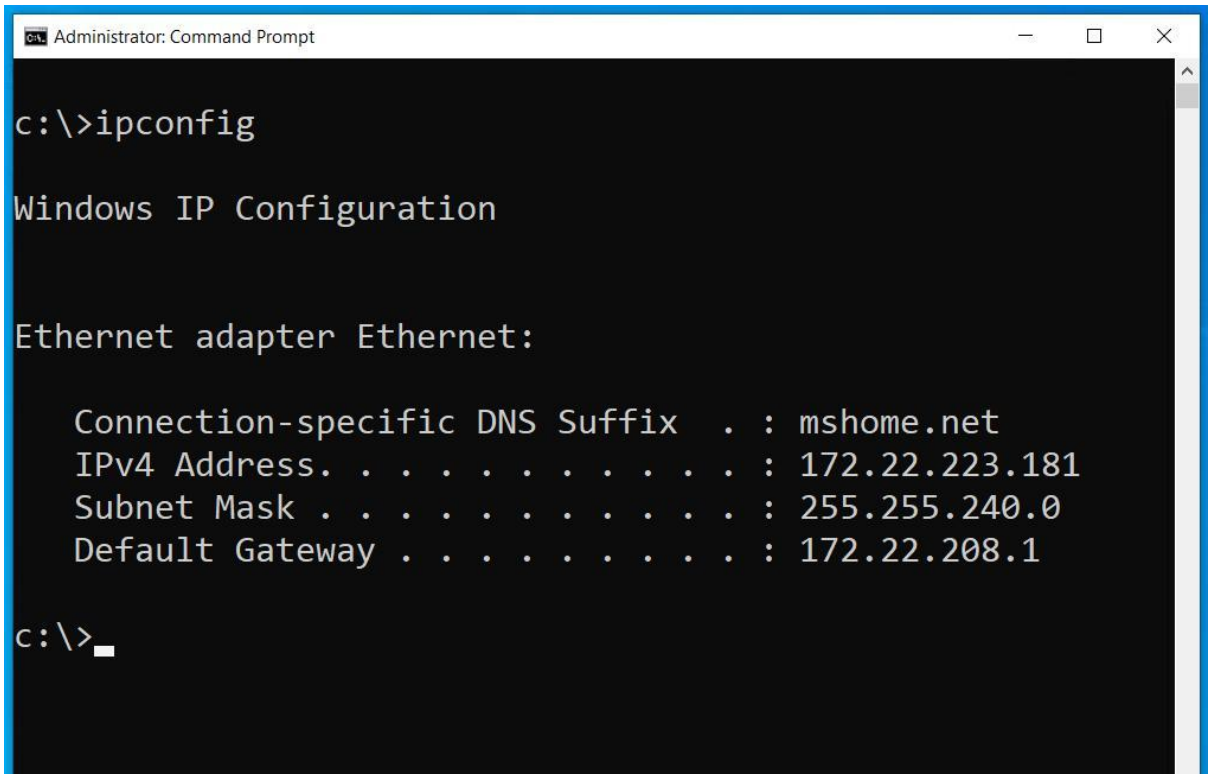
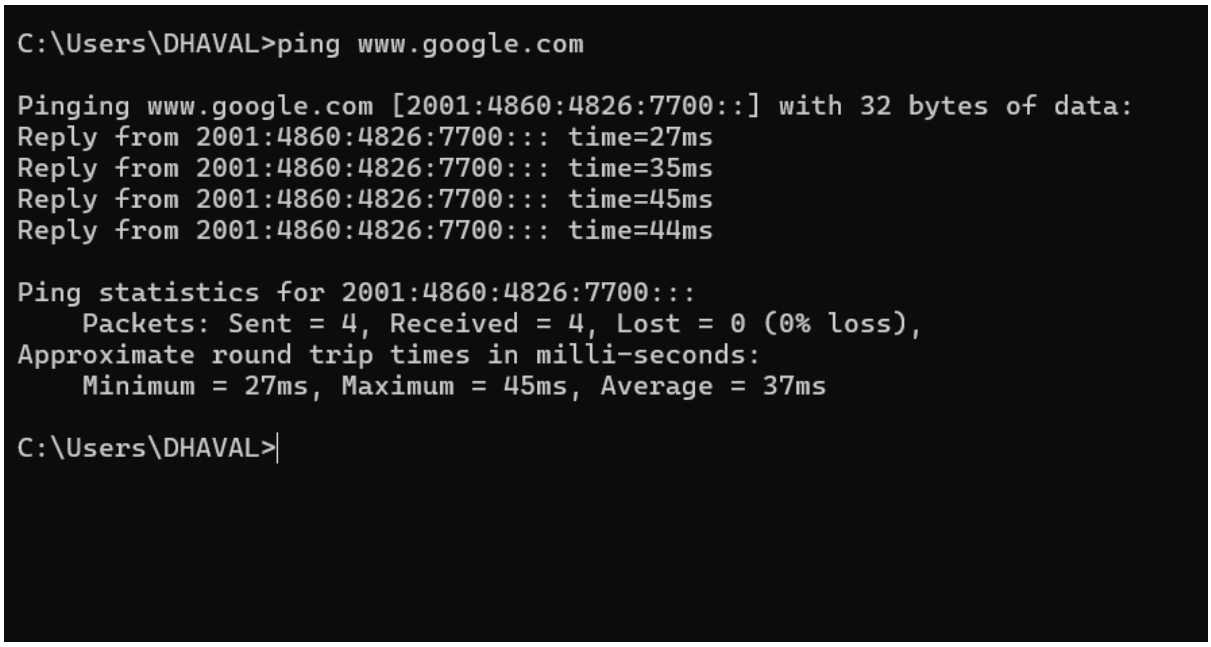
Assignment – 2.6

Troubleshooting Tools

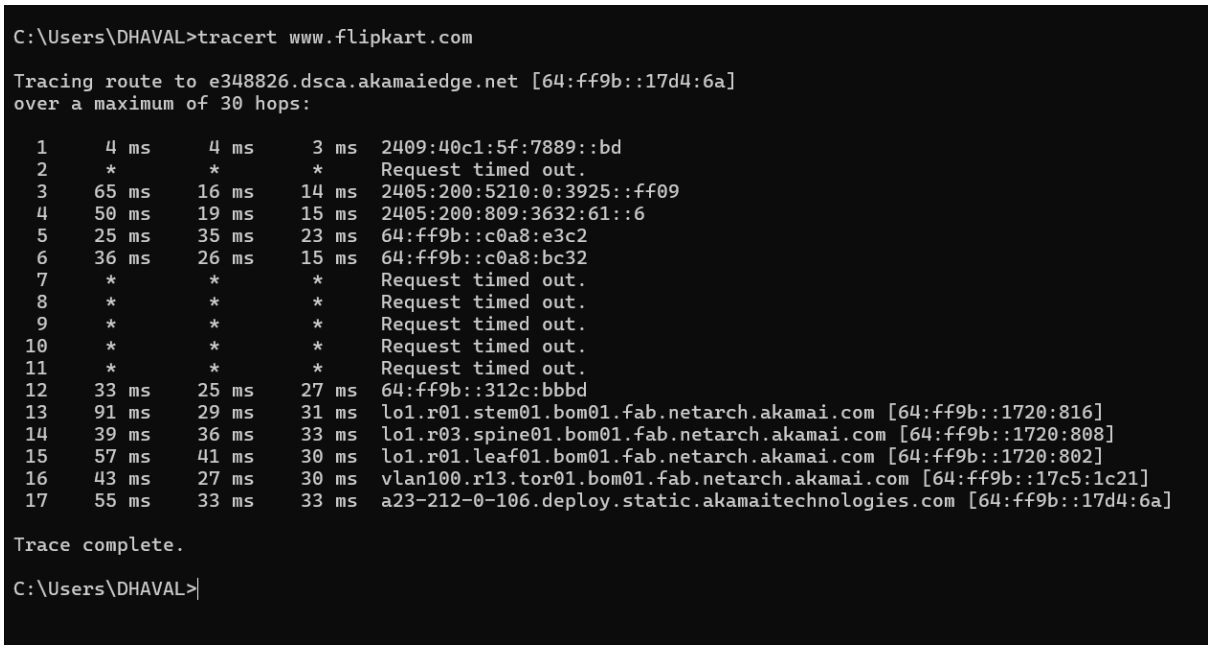
Q - 1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



<u>Q - 2.</u> Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.



Q - 3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.



```
C:\Users\DHAVAL>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [64:ff9b::17d4:6a]
over a maximum of 30 hops:

  1     4 ms     4 ms     3 ms  2409:40c1:5f:7889::bd
  2     *        *        *     Request timed out.
  3    65 ms    16 ms    14 ms  2405:200:5210:0:3925::ff09
  4    50 ms    19 ms    15 ms  2405:200:809:3632:61::6
  5    25 ms    35 ms    23 ms  64:ff9b::c0a8:e3c2
  6    36 ms    26 ms    15 ms  64:ff9b::c0a8:bc32
  7     *        *        *     Request timed out.
  8     *        *        *     Request timed out.
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11     *        *        *     Request timed out.
 12    33 ms    25 ms    27 ms  64:ff9b::312c:bbbd
 13    91 ms    29 ms    31 ms  lo1.r01.stem01.bom01.fab.netarch.akamai.com [64:ff9b::1720:816]
 14    39 ms    36 ms    33 ms  lo1.r03.spine01.bom01.fab.netarch.akamai.com [64:ff9b::1720:808]
 15    57 ms    41 ms    30 ms  lo1.r01.leaf01.bom01.fab.netarch.akamai.com [64:ff9b::1720:802]
 16    43 ms    27 ms    30 ms  vlan100.r13.tor01.bom01.fab.netarch.akamai.com [64:ff9b::17c5:1c21]
 17    55 ms    33 ms    33 ms  a23-212-0-106.deploy.static.akamaitechnologies.com [64:ff9b::17d4:6a]

Trace complete.

C:\Users\DHAVAL>
```

<u>Q - 4.</u> Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.



Q - 5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check. Hint: Think about checking both network connectivity and DNS resolution.

- **Troubleshooting commands to diagnose Zomato website access issue:**

  1. **ipconfig:** Checks the laptop's IP address, subnet mask, and default gateway to verify that the network configuration is correct.

  2. **ping:** Run ping www.zomato.com to check whether the laptop can communicate with Zomato's server and measure response time.

  3. **nslookup:** Run nslookup www.zomato.com to check whether DNS is working correctly and resolving Zomato's domain name into an IP address.